A Case Study of Microplastics Detection through an Integrated STEM Framework

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***Abstract*—Microplastics pollution is a severe threat to ecosystems and human health. Unfortunately, traditional spectroscopic identification methods are still cost-prohibitive for the majority of undergraduate laboratories. This paper introduces a cross-disciplinary integrated STEM (iSTEM) framework, which transforms students from passive learners to active environmental researchers through a four-phase pipeline process. This includes wet-lab density separation, low-cost USB microscopy imaging, the deployment and comparative evaluation of 4 deep learning architectures (Faster R-CNN, YOLOv5, U-Net, and Mask R-CNN) and a unified bounding-box evaluation methodology enabling direct cross-paradigm comparison. Optical artifacts synthesis incorporated by a three-stage training sequence to few-shot domain adaptation. Mask R-CNN obtained the highest balanced performance (F_1=0.7978), while YOLOv5 maximized recall and U-Net maximized precision (0.8619/0.7546). All models proved to statistically significantly exceed random-chance baseline ($p<0.05$). The obtained results provide evidence of the assumption that deep learning tools can be effectively applied in resource-constrained educational settings providing a replicable blueprint for the proposed iSTEM educational integration.**

Keywords—microplastics, image segmentation, feature extraction, deep learning, environmental monitoring, STEM education, interdisciplinary learning, low-cost workflows.

I. Introduction

Microplastics—plastic particles smaller than 5 mm—are ubiquitous contaminants found in oceans, rivers, sediments, soils, and biological tissues. Their ingestion by aquatic organisms and humans leads to bioaccumulation, posing significant ecological and health risks. Detecting and quantifying these particles is essential for understanding their environmental and physiological impacts. Microplastics have been recognized as being a serious problem in both ecological systems and human health and they are found in various environmental compartments [1], [2]. Recent toxicological reviews further emphasize that these particles can disrupt cellular functions and trigger immune responses [3].

Regarding chemical recognition, spectroscopic methods such as SERS were heavily tested [4]. Application of deep learning models, such as U-Net++ [5], is increasingly relevant not only in pure research but also in interdisciplinary E-STEM educational frameworks [6]. Furthermore, advanced architectures like Mask R-CNN [7] and rapid single-stage detectors like YOLOv5 [8] have been deployed for microplastics segmentation and detection with great success. The necessity for these automated detection tools is underscored by the fact that microplastics are vastly more prevalent in oceanic habitats than historically assumed [9].

The most commonly used methods for microplastics identification are Fourier-transform infrared spectroscopy (FTIR) and Raman spectroscopy that are highly dependable but costly and time-consuming, whereas manual visual microscopy, which is characterized by ease of use, is subjective, tedious and error-prone [10]. Prior to any visual or computational analysis, standard laboratory extraction methods using density separation and filtration are required to isolate these particles from environmental samples [11].

Recent advances in image processing and deep learning have enabled automated detection and classification of microplastics. The initial research focused on using normal shape and texture descriptors for microplastics classification, like Local Binary Patterns (LBP) [12] and morphological features, alongside other advanced spectroscopic techniques to identify microplastics in aquatic environments [13]. Building upon these foundations, Convolutional Neural Networks (CNNs) have been widely adopted to automatically categorize microplastic forms [14]. Other specialized architectures, such as Faster R-CNN, have successfully mapped specific polymer types [15]. Consequently, there is an increasing consensus on the critical need for standardized machine learning frameworks dedicated to environmental monitoring [16].

Despite these scientific advances, deploying such models in low-cost laboratory workflows suitable for novice students remains a significant challenge. Furthermore, paradigm-specific evaluation metrics (e.g., mean IoU versus mean average precision) make direct model comparison difficult. This study overcomes that challenge from an integrated STEM (iSTEM) perspective, framing the work specifically to educate iSTEM competencies. Prioritizing teaching effectiveness over architectural novelty, this study introduces a common bounding-box evaluation pipeline, allowing both object detection and segmentation models to be compared under identical low-cost conditions.

II. Literature Review

A. Environmental and Human Health Impacts

The lack of urgency and neglect towards environmental and health risks of microplastic pollution is the reason why this research has been initiated. Due to their small size and potential dangers in waste disposal, microplastics play an integral role in the transport of organic and inorganic contaminants in aquatic environments. Being the most

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significant source of bioaccumulated contaminants, microplastics have high surface-area-to-volume ratios and hydrophobic properties for heavy metals and persistent organic pollutants (POPs) [1]. Moreover, the other evidence of the effects on human health is a 2024 cardiovascular study, Marfella *et al* [2], which reported that microplastics detected in carotid artery plaques of patients had a 4.53-fold greater risk of myocardial infarction, stroke, or of all-cause mortality. Further emphasize the findings from a 2025 review in Frontiers in Public Health which stated that microplastic particles, by disrupting mitochondrial energy production, promote pro-inflammatory cytokine release and trigger the immune system [3]. To address these pressing health and ecological threats, the scientific community has rapidly advanced detection methodologies; for instance, chemical recognition techniques like Surface-Enhanced Raman Scattering (SERS) have been deployed to trap and identify these tiny pollutants [4], while deep learning architectures such as U-Net++ have established highly effective baselines for complex segmentation tasks [5].

B. Experiential Learning in iSTEM

Constructivist learning theory emphasizes active learner participation. Integrating environmental education with STEM (E-STEM) effectively bridges theoretical knowledge and real-world ecology to raise environmental awareness [6]. Aligning with this, our project-based iSTEM curriculum transforms abstract pollution concepts into tangible experiences. By engaging in hands-on pollutant extraction and computational analysis, students transition from passive observers to active environmental contributors. To facilitate this computational analysis, Cabaneros et al. [7] utilized enhanced U-Net and Mask R-CNN models to track environmental debris under poor resolution. Additionally, Arju et al. [8] successfully deployed YOLOv5 architectures for the low-cost detection of plastic waste using mobile microscopy.

C. Environmental and Laboratory Extraction Methods

Thompson et al. [9] are among the first researchers to explicitly recognize microplastics as distinct pollution agents, stating they are vastly more prevalent in fragile oceanic habitats than initially thought. Thereafter, Hidalgo-Ruz et al. [10] reviewed the various identification methods of microplastics, but one striking fact they emphasized was that there were absolutely no universally standard protocols during the initial sampling period, which is a major limiting factor that negatively affects accurate comparability among global studies. The method that is most practical for students is the one by Masura et al. [11], which utilizes the density mobility of sodium chloride (NaCl) solutions and filtration. Besides, this is still the best practical method to isolate microplastics from raw environmental samples in resource-poor conditions. By participating in a project that tackles actual environmental pollution using easy-to-find household materials, students not only learn foundational concepts such as specific gravity, buoyancy, and density separation in chemistry, but they are also directly involved in the practical application of these concepts to solve urgent ecological problems. This methodology effectively bridges the critical gap between theoretical classroom science and actual environmental stewardship.

D. Image Analysis and Feature Extraction for Microplastics

Manual microscopy is time-consuming, subjective, and highly prone to visual fatigue. Early automated analysis relied on shape and texture descriptors like Local Binary Patterns [12], while spectroscopic techniques like surface-enhanced Raman scattering [13] enabled chemical identification. Deep learning accelerated this progress, with Convolutional Neural Networks (CNNs) significantly outperforming prior categorization methods [14]. Furthermore, architectures like Faster R-CNN have successfully mapped specific polymers [15], highlighting the growing need for standardized environmental machine learning frameworks [16]. However, integrating these AI advances into undergraduate curricula is hindered by the lack of unified benchmarking for low-resource contexts. This study addresses this gap by introducing a comparative teaching methodology to evaluate multiple AI models under identical, low-cost conditions.

III. METHODOLOGY

What is unprecedented in this study is that it has the methodical translation of the complex, and very high-level artificial intelligence research into an easily understandable, and hands-on full laboratory teaching method. In contrast to regarding AI as a "black box," the explicit main feature of this method is to make a step by step guide to students through four fundamental elements of iSTEM: Chemical Sciences (wet-lab extraction), Hardware Engineering (budget constraint management), Technological Application (AI deployment), and Mathematical Unification (standardized evaluation).

A. Phase 1: Chemical and Physical Sciences (Wet-Lab Extraction)

The procedural steps described in the laboratory protocol identify it as a hands-on learning environment that seeks to maximize student participation in the physical sciences:

- Sample Collection: The environmental samples were taken directly from Bang Saen beach at Chonburi, Thailand. This helps students utilize a popular local coastline to bridge the gap between abstract text in the textbook and pollution, and their immediate reality.
- Density Separation: A saturated NaCl solution is prepared by mixing equal parts of salt and water so that addition of salt makes the solution unsaturated at high temperature. The strained sample is then added to the solution of saltwater. The solution is stirred and left untouched for 3 to 4 hours to fully settle. In this case, students are introduced to the chemical principles of specific gravity and buoyancy.
- Filtration and Transfer: The upper layer containing the lighter plastic particles is separately filtered through a fine mesh sieve with caution. The collected material is air-dried in an area that is isolated from wind contamination before it is transferred to a clean surface.
- Non-Destructive Confirmation: The visual examination under controlled illumination tests the translucency of particles. The choice of non-destructive optical confirmation was made specifically over the old-fashioned hot-needle test to keep the sample intact for the further computational

stages. It should be noted that this visual confirmation method does not constitute chemical validation of polymer identity (e.g., via FTIR or Raman spectroscopy); this limitation is discussed in the Limitations section.

B. Phase 2: Hardware Engineering (Low-Cost Digitization)

Imaging utilized a consumer-grade 40x USB digital microscope with dual LED illumination, capturing 4032×3024 images resized to 1920×1080 for processing. These low-cost optics introduced domain-specific noise—such as optical glare and chromatic aberration—requiring computational mitigation by students. Samples on standard glass slides were digitized and manually annotated via Roboflow. The 40x magnification yields a minimum detection limit of ~0.6 micrometers; a limitation perfectly aligned with the framework's educational objectives, which deliberately prioritizes accessibility over extreme analytical sensitivity.

C. Phase 3: Technological Application (AI Deployment)

Four different pretrained deep learning models were evaluated as distinct computer vision paradigms: Faster R-CNN and YOLOv5 (object detection), and U-Net and Mask R-CNN (semantic and instance segmentation, respectively).

In order to cope with the teaching issue of "domain shift" that was encountered due to the cheap USB microscope, the approach introduces a completely new, three-phase student-driven training sequence:

- Sub - Phase 1 (Initial Knowledge Training): The models were trained with the public "panats-mp-project" dataset to be able to extract basic shape and edge features.
- Sub - Phase 2 (Environmental Fine-Tuning): The models were fine-tuned with 50 locally taken microscopy images of actual microplastics collected from Bang Saen beach sand.
- Sub - Phase 3 (Domain Adaptation): The last set of data consisted of 132 strictly regulated images. Out of these, 30 images were chosen to make a precise subset for the few-shot domain adaptation phase which was carried out to help the models mathematically learn the specific lighting artifacts of the microscope environment. The other 102 images that were left out were used for final hypothesis testing.

D. Phase 4: Mathematical Unification (Evaluation Pipeline)

Object detection models (Faster R-CNN, YOLOv5) natively produce bounding boxes, while segmentation models (U-Net, Mask R-CNN) output pixel-level masks, complicating direct comparison. To enable cross-paradigm evaluation, segmentation outputs were post-processed into standard bounding boxes. Morphological Opening (erosion followed by dilation) removes noise, while Closing (dilation followed by erosion) connects fragmented predictions. Finally, contour detection extracts the minimum enclosing rectangle for each mask. This unified scheme ensures performance variations reflect only architectural differences.

Pedagogical Justification: This unified bounding-box evaluation was adopted as a deliberate pedagogical design choice. We acknowledge that this conversion neutralizes the intrinsic spatial precision of segmentation models, which would be better captured by native metrics (e.g., mIoU, Dice coefficient). However, this trade-off was deemed acceptable for the educational objective of enabling accessible cross-model comparison under identical conditions.

E. Contamination Severity Quantification (C_p)

In order to turn the raw AI detection outputs into usable environmental metrics, the framework proposes a unique teaching computation called contamination percentage (C_p). For each image, the cumulative area of all detected bounding boxes is computed relative to the total image area. A cap at 100% prevents overflow from overlapping detections:

$$C_p = \min\left(100.0, \left(\frac{\sum Area(B_{\mathrm{i}})}{Area(I_{\mathrm{total}})}\right) \times 100\right), \quad (1)$$

where $\sum Area(B_{\mathrm{i}})$ is the cumulative area of all positive bounding boxes, and $Area(I_{\mathrm{total}})$ is the total pixel resolution of the microscopic image. This mathematical step explicitly links computer science to environmental toxicology severity.

F. Performance Metric Calculation

A predicted bounding box was classified as a true positive (TP) if its Intersection over Union (IoU) with a ground-truth annotation exceeded a threshold of 0.5. Otherwise, it was classified as a false positive (FP). Ground-truth annotations not matched by any prediction were classified as false negatives (FN). True negatives are not defined in object detection tasks.

Precision was calculated to measure how many detected objects were actually microplastics: Precision = TP / (TP+FP).

Recall was calculated to measure how many of the actual microplastic particles were detected: Recall = TP / (TP+FN).

To provide a balanced evaluation between precision and recall, the F_1-score was computed as follows:

$$F_1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}). \quad (2)$$

The identification of a true positive was validated using the Intersection over Union (IoU) metric against the ground-truth annotations:

$$IoU = \frac{Area(B_p \cap B_g)}{Area(B_p \cup B_g)}. \quad (3)$$

Based on IoU matching results, each prediction was categorized as a true positive (TP), false positive (FP), or false negative (FN). Precision, Recall, and the F_1-score were calculated uniformly across all four architectures.

IV. Results and Discussion

A. Quantitative Performance Evaluation

Following few-shot domain adaptation, all models showed substantial performance improvements. Mask R-CNN achieved the highest balanced performance, with an F_1-score balance of 0.7978. Background noise was a minimal factor; thus, the instance segmentation backbone acted like a magic pencil to solve the low-budget microscopy noise issue while still achieving an excellent precision of 0.7725 and a recall of 0.8249. Faster R-CNN was a two-stage detector that reliability

extensively performed the task and getting second position with an F_1-score of 0.7425 (Table I).

YOLOv5 and U-Net illustrate a clear sensitivity-specificity trade-off. YOLOv5 achieved the highest recall (0.8619) but its single-stage design struggled to suppress background artifacts, yielding the lowest precision (0.6215). Conversely, U-Net's morphological post-processing effectively eliminated false positives for high precision (0.7546), but this conservative approach missed smaller particles, lowering recall (0.6949).

TABLE I. COMPARISON OF EVALUATED MODELS.

Model	Precision	Recall	F_1-Score
Mask R-CNN	0.7725	0.8249	0.7978
Faster R-CNN	0.7135	0.7740	0.7425
U-Net	0.7546	0.6949	0.7235
Yolov5	0.6215	0.8619	0.7222

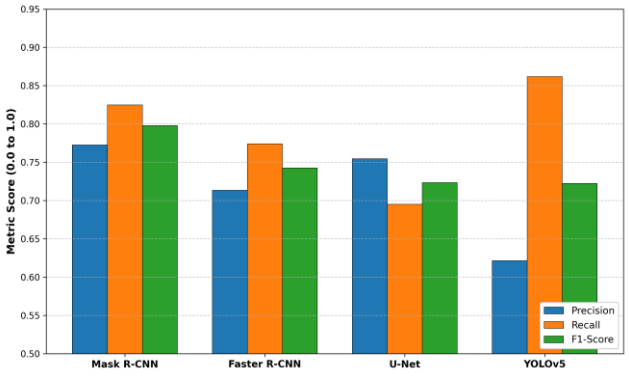


Fig. 1. Grouped bar chart comparing the Precision, Recall, and F_1-Score of the evaluated AI architectures, clearly illustrating the sensitivity-specificity trade-off.

These trade-offs provide a valuable pedagogical outcome: students learn that YOLOv5 functions as a high-sensitivity screening tool, U-Net serves as a high-specificity filter, and Mask R-CNN offers the best overall balance for general-purpose environmental monitoring.

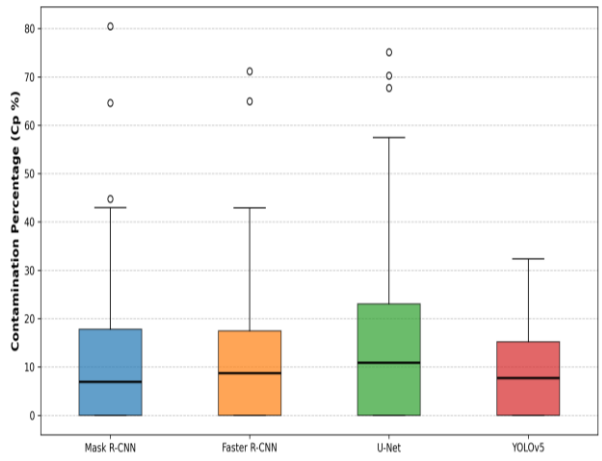


Fig. 2. Cross-architectural Boxplot demonstrating the contamination percentage (C_p) distribution.

The contamination percentage (C_p) metric facilitated not simply ranking but also the quantification of environmental contamination. Measuring this metric on the holdout dataset of Bang Saen beach, the system demonstrated the spread of the optical pollution through all the 102 samples.

Figure 2 presents boxplots of the Cp distributions for each model. U-Net and Faster R-CNN produced stable, centralized distributions with medians between 8% and 11%.

B. Qualitative Visualization Results and Image Placement Guidelines

1) Faster R-CNN Results

Faster R-CNN demonstrated reliable detection of microplastic particles across varied shapes and sizes, with strong spatial agreement between predicted and ground-truth bounding boxes (see Fig. 3).



Fig. 3. Faster R-CNN detection results showing strong agreement between ground truth (green) and predicted bounding boxes (red) for microplastic particles across varied shapes and sizes.

2) YOLOv5 Detection Results

YOLOv5 achieved high-recall detection but produced notable false positives in low-contrast regions where background dust and optical artifacts were misclassified as microplastics (see Fig. 4).



Fig. 4. YOLOv5 detection results illustrating high-recall detection of microplastic particles, with notable false positives from background dust and optical artifacts in low-contrast regions.

3) U-Net Semantic Segmentation Results

U-Net produced the most precise spatial boundaries through morphological post-processing, with substantially fewer false positives than the single-stage detectors (see Fig. 5).

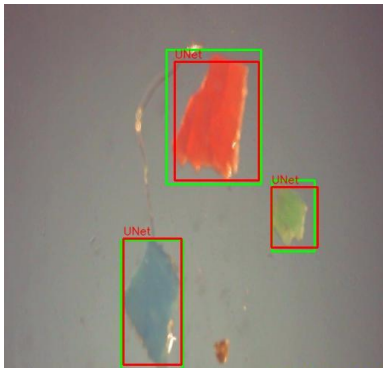


Fig. 5. U-Net semantic segmentation results demonstrating precise spatial boundaries achieved through morphological post-processing, with reduced false positives compared to single-stage detectors.

4) Mask R-CNN Instance Segmentation Results

Following domain adaptation, Mask R-CNN showed the strongest overall balance between mask accuracy and bounding-box precision, with effective suppression of background noise (see Fig. 6).

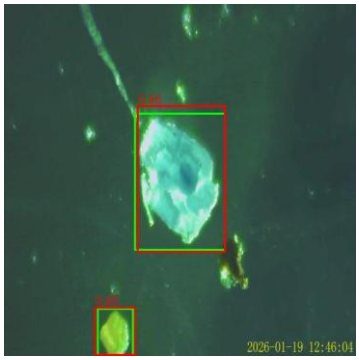


Fig. 6. Mask R-CNN instance segmentation results showing the best overall balance between mask accuracy and bounding box precision, with effective suppression of background noise following domain adaptation.

V. Educational Outcomes and Reflective Practice

This section evaluates the pedagogical effectiveness of the iSTEM framework through reflective practice narratives of the first three co-authors (undergraduate information technology students) in developing and implementing the pipeline, combined with a structured mapping of learning outcomes to framework phases. Although the small sample size (n = 3) limits generalizability, these reflections provide qualitative evidence of the framework's capacity for cross-disciplinary learning.

A. Navigating Cross-Disciplinary Challenges

For the first three co-authors, the information technology (IT) coursework had been limited to software engineering and computer science. The iSTEM framework required a direct engagement with wet-lab chemistry, specific gravity calculations, and filtration techniques. The optical imaging phase served as a reality check on hardware limitations: unlike the curated digital datasets IT students are accustomed to, the low-cost microscope produced significant optical glare. A fundamental engineering principle, that computational models are only as reliable as the data they receive, was taught by bridging the gap between physical sample preparation and digital analysis.

B. The Tangibility of the Local Health Threat

The microplastics health crisis was an abstract concept that existed only in the minds of people and the textbooks they read before the initiation of the project. The process of domestic sample examination from Bang Saen beach such as detecting polymer fragments with the microscope and pollution percentage calculation inverted a global problem to a local one. This kind of practical work was the main nurturing factor of the environmental awareness that only the classical courses could never reach.

VI. Statistical Validation of Microplastics Detection

In order to prove that our models only detect the real microplastics and not the random artifacts we have performed the hypothesis testing employing a right-tailed one-sample z-test against the random-chance baseline of 50% (p_0 = 0.50$). The instances that were measured (n) are the total of True Positives, False Positives, and False Negatives collected from the testing stage.

The statistical significance of each model's detection capability was calculated using the following z-test formula for a single proportion:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}, \tag{4}$$

where $\hat{p}$ is the observed classification accuracy (True Positives divided by n), p_0 is the hypothesized random-chance baseline (0.50), and n is the total number of evaluated instances.

Crucially, the testing dataset included negative controls consisting of crushed glass particles. Therefore, the False Positives recorded in this statistical evaluation primarily represent instances where the models were optically "tricked" by the translucent properties of the glass.

Because we are strictly evaluating whether the models' accuracy is significantly greater than random chance, this was formulated as a right-tailed test. For all four architectures, the null hypothesis ($H_0: p \leq p_o$) states that the model cannot reliably detect real microplastics and its classification accuracy is equal to or worse than random chance. The alternative hypothesis ($H_a: p > p_0$) states that the models can reliably differentiate microplastics from background noise and negative glass controls at a rate significantly higher than random chance.

TABLE II. *Z*-Test Results for Hypothesis Testing of Microplastics Detection Models.

Model	n	z - value	p -value
Mask R-CNN	220	4.8542	0.001
Faster R-CNN	232	2.7574	0.0029
YOLOv5	276	2.1669	0.0151
U-Net	217	1.9686	0.0245

The null hypothesis got rejected for all four models with a significance level of α = 0.05 as we can clearly observe in Table II. Among them, Mask R-CNN had the highest statistical significance (z = 4.8542, p = 0.001), achieving an overall instance accuracy of 66.36%. Similarly, Faster R-CNN also showed a very strong statistical dependence (z = 2.7574, p = 0.0029). It is worthy of note that although YOLOv5 recorded the highest number of instances (n = 276) among all models, mainly due to its high false-positive rate on the glass

controls and background noise, it still managed to achieve a net accuracy of 56.52%, which was statistically significant (z = 2.1669, p = 0.0151). Lastly, U-Net, which recorded the fewest overall instances (n = 217) as a result of its conservative thresholding, was also successful in reaching the significance cut-off (z = 1.9686, p = 0.0245). These results confirm that both single-stage detectors and segmentation models can reliably detect microplastics when used with a unified bounding-box evaluation pipeline and few-shot domain adaptation.

The test statistic, which is the z-test against a 50% random baseline, has been selected for its handiness in the context of undergraduate education. While this method is unconventional in computer vision research, where mean average precision (mAP at IoU 0.5 and mAP at IoU 0.5:0.95) serves as the standard evaluation metric for object detection, it gives a means to the students to use an easy statistical framework to verify that the model performs better than the random scheme. In this respect, it should be borne in mind that these data should be seen only as an additional verification and not as a replacement of standard detection benchmarks.

VII. Limitations and Future Work

The most significant restrictions are those associated with this research. To begin with, polymer fragments were labeled according to their optical translucency as observed with the help of a microscope instead of a chemical validation (e.g., FTIR or Raman spectroscopy), which has a poor performance in telling the difference between polymer fragments and organic debris. Therefore, there was a potential label noise. Secondly, the 40× USB microscope was operating on a minimum threshold of 8 micrometers but it could only detect particles that were larger than that (upper microplastics fraction). Third, the bounding-box evaluation of the whole has not captured the segmentation models full pixel-level precision; native metrics mIoU and Dice coefficients should be reported in future works.

The few-shot domain adaptation of the educational assessment is not benchmarked against other alternatives such as contrast limited adaptive histogram equalization (CLAHE) or standard data augmentation strategies. Additionally, the evaluation is only of reflective narratives from the first three co-authors (n = 3) as IT students, thereby creating a precedent case study. In future implementations, an introduction of pre/post-intervention surveys and rubric-based evaluation would lead to a more robust claim regarding pedagogy.

VIII. Conclusion

The outlined case study displays a practical example of the first three co-authors as undergraduate IT students having all the variations: building, testing, and completing an environmental monitoring pipeline - a hardware-independent AI in a real low-cost project using commercial off-the-shelf (COTS) tools. The four-phase iSTEM framework consists of wet-lab chemistry, hardware-imaging, deep learning and final statistical evaluation which is a means to integrate two distinct subjects - computational engineering and environmental science. Mask R-CNN was the detection that reached the most suitable trade-off between accuracy and recall (F_1 = 0.7978), while YOLOv5 maximized sensitivity, and U-Net exhibited the highest precision. As a whole, all four frameworks would have dispelled random-chance baselines completely if deep learning could have been used in financially limited conditions. This study was initiated from local environmental data to address the problem of microplastics pollution from the perspective of data science, thus creating a sample action plan for other tutors interested in merging computing education with environmental science.

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